

CAT 2025

MBA FASTRACK

Lecture - 03

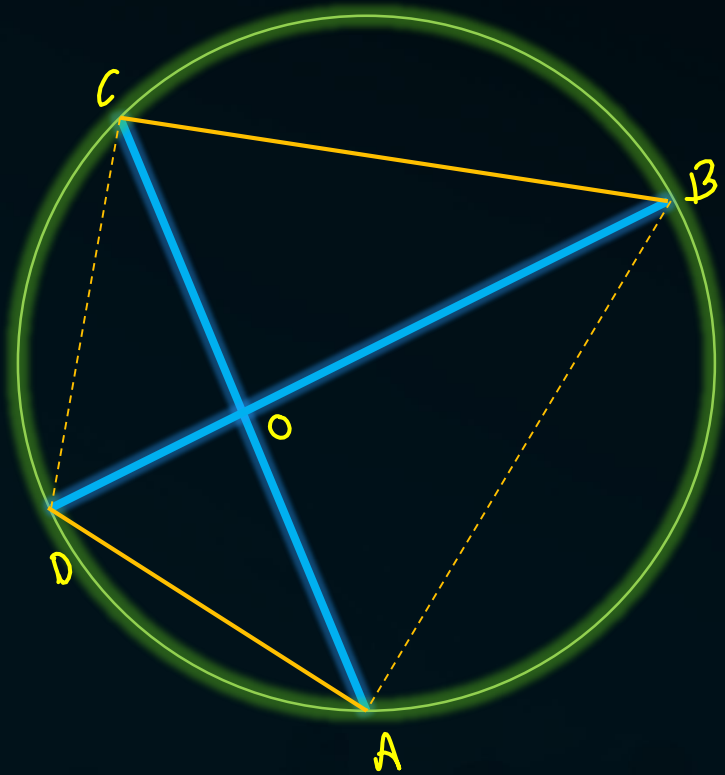
Geometry 3

Circles – Quads – Polygons – 3D

By- RAHUL BATHLA



RECAP OF THE PREVIOUS LECTURE



$$(DO)(BO) = (AO)(CO)$$

$$\frac{CO}{AO} = \frac{(CO)(CB)}{(AO)(AB)}$$

$$\frac{BO}{DO} = \frac{(BC)(BA)}{(DC)(DA)}$$

TOPICS *to be covered*



Topic

Circles Remaining

Topic

Quadrilaterals

Topic

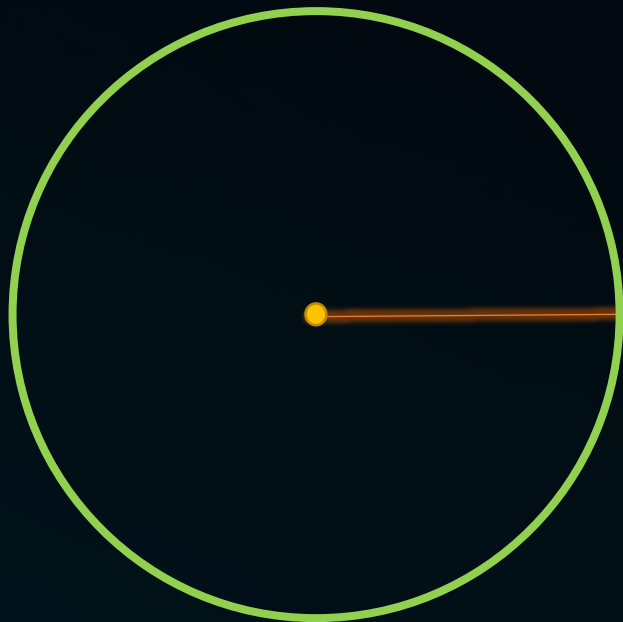
Polygons

Topic

Surface Area + 3D

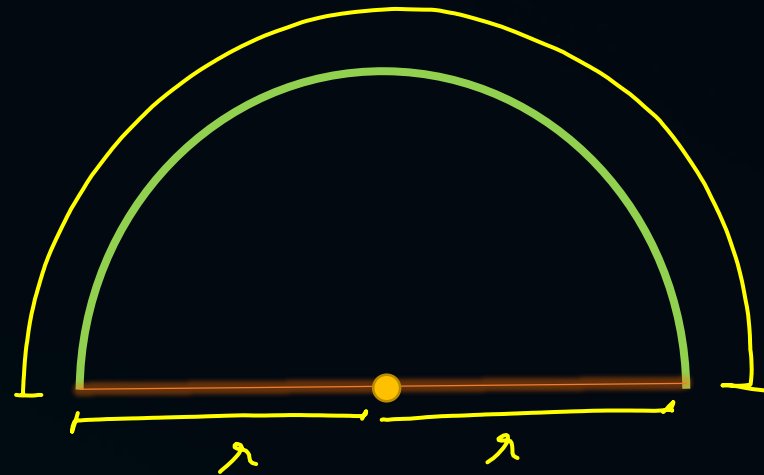


Topic: Circles Remaining



$$\text{Area} = \pi r^2$$

$$\text{Perimeter} = 2\pi r$$



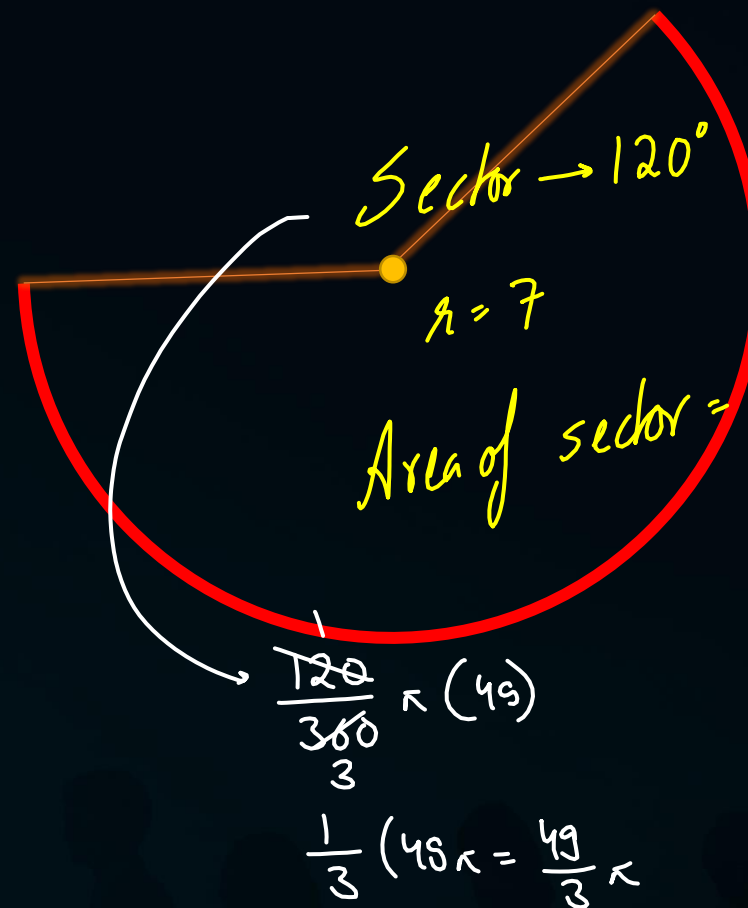
$$\text{Area} = \frac{\pi r^2}{2}$$

$$\text{Perimeter} = \pi r + s$$

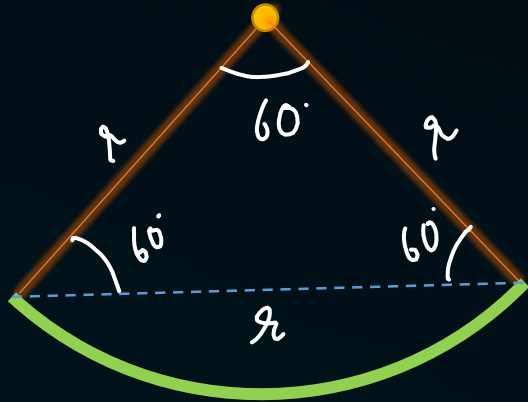


$$\text{Area} = \frac{\theta}{360} (\pi r^2)$$

$$\text{Perimeter} = \frac{\theta}{360} (2\pi r) + 2r$$

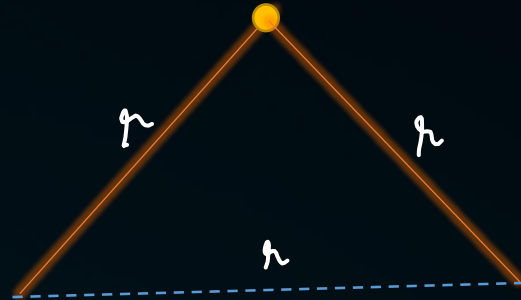


Sector and Segment (Centre Angle 60 degrees)



$$\text{Area} = \frac{1}{6} \pi r^2$$

$$P = \frac{1}{6} (2\pi r) + 2r$$



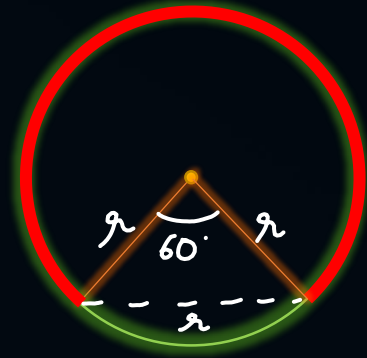
$$\text{Area} = \frac{\sqrt{3}}{4} r^2$$



SEGMENT

$$\text{Area} = \frac{1}{6} \pi r^2 - \frac{\sqrt{3}}{4} r^2$$

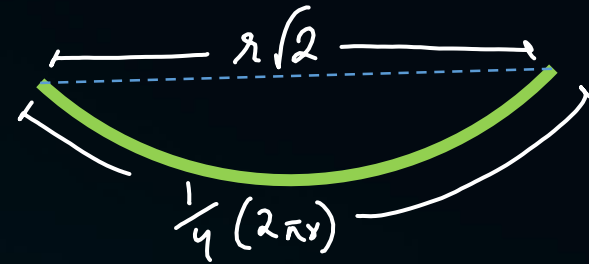
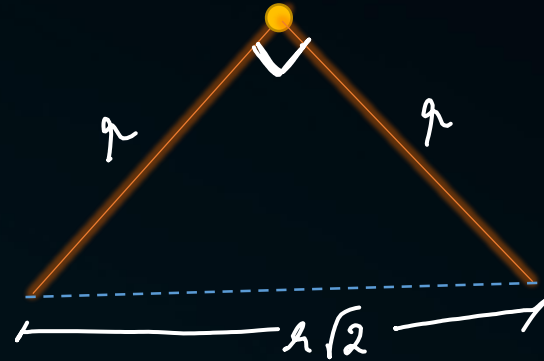
$$\text{PERIMETER} = \frac{1}{6} (2\pi r) + r$$



(CHORD LENGTH
= RADIUS)

SECTOR WITH
CENTRE ANGLE
= 60°

Sector and Segment (Centre Angle 90 degrees)



$$\text{Area} = \frac{1}{4} \pi r^2$$

$$\text{Area} = \frac{1}{2} r^2$$

$$P = \frac{1}{4} (2\pi r) + \sqrt{2} r$$

$$A = \frac{1}{4} \pi r^2 - \frac{1}{2} r^2$$

QUADRANT
OF CIRCLE
CENTRE ANGLE
= 90°

$$P = \frac{1}{4} (2\pi r) + 2r$$

QUESTION - 1



#Q. If the area of a first Circle is equal to the sum of the area of the other two circles having the diameters as 10cm and 24 cm. Find the radius of the first circle.

- A** 15 cm
- B** 13 cm ✓
- C** 12 cm
- D** 10 cm

$$\pi R^2 = \pi(5)^2 + \pi(12)^2$$

$$R^2 = 25 + 144$$

$$R^2 = 169$$

$$R = 13 \checkmark$$

QUESTION – 2



#Q. In a circle with center 'O', AB is a chord which is equal to the radius of the circle. Find the Area of the Minor segment formed by AB and the circle given that the radius of the circle is 6cm.

A

$$3 (\pi - \sqrt{3})$$

B

$$2 (3\pi - 3\sqrt{3})$$

C

$$3 (2\pi - 3\sqrt{3})$$

D

$$3 (\pi + \sqrt{3})$$

$$\frac{1}{2} (\pi) (36) - \frac{\sqrt{3}}{4} 36$$

MINOR SEGMENT

$$6\pi - 9\sqrt{3}$$

$$\begin{aligned} \text{MAJOR} &= \text{CIRCLE} - \text{MINOR} \\ &= \pi (36) - [6\pi - 9\sqrt{3}] \\ &= 30\pi + 9\sqrt{3} \end{aligned}$$



QUESTION – 3 (CAT 2019)



#Q. Two circles, each of radius 4 cm, touch externally. Each of these two circles is touched externally by a third circle. If these three circles have a common tangent, then the radius of the third circle, in cm, is

A

$\pi/3$

B

1

C

$1\sqrt{2}$

D

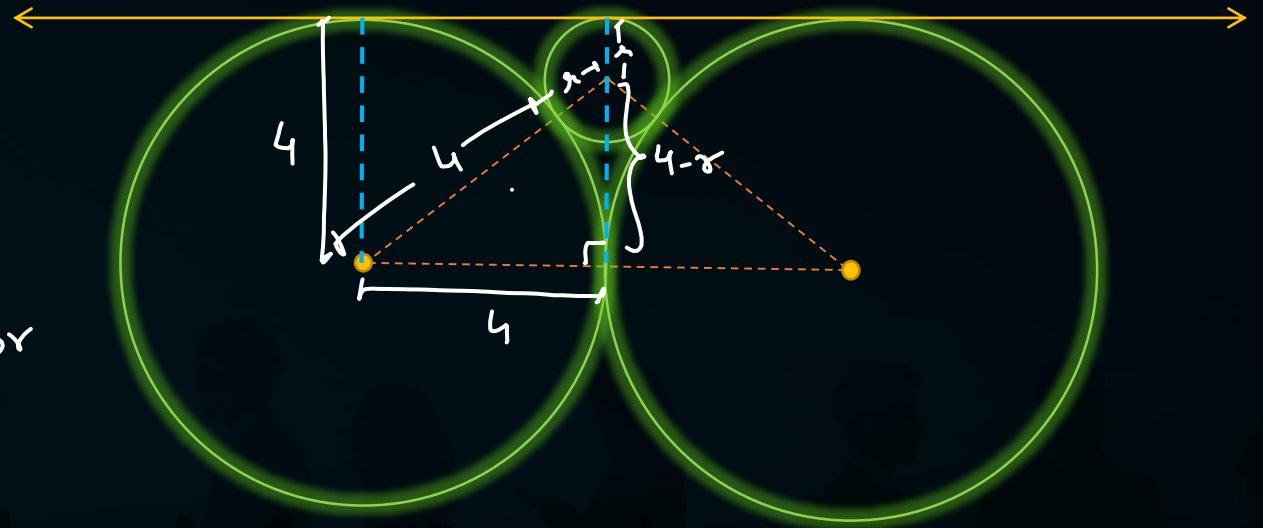
$\sqrt{2}$

$$(4+r)^2 = 4^2 + (4-r)^2$$

$$16 + r^2 + 8r = 16 + 16 + r^2 - 8r$$

$$16r = 16$$

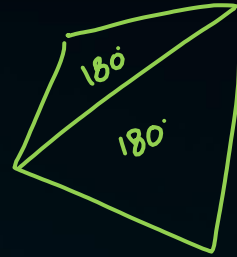
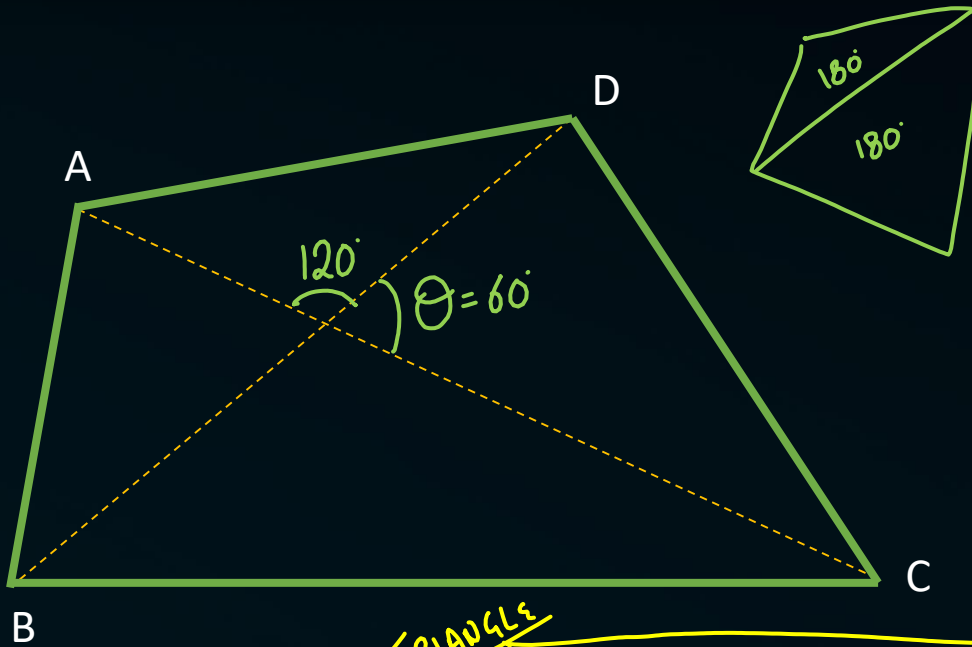
$$r = 1$$





Topic: Quadrilateral

Quadrilaterals



- A quadrilateral is a plane figure enclosed by four edges (sides) and four vertices.
- The line joining the opposite vertices is called diagonal.
- Sum of the angles of quadrilateral = 360°

TRIANGLE

$$3 < c < 7$$

$$2, 5, c$$

$$2, 3, 7, c$$

❖ Sum of 3 sides of a Quadrilateral is always greater than the fourth side.

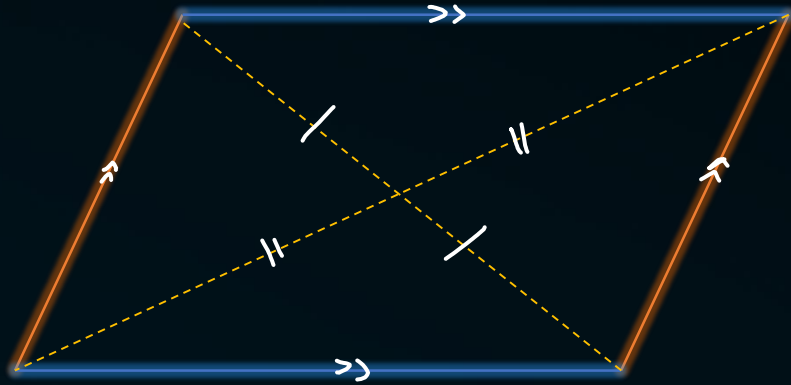
❖ Area of any quadrilateral = $\frac{1}{2} (AC)(BD) \sin Q$

$$7 - (2 + 3) < c < 12$$

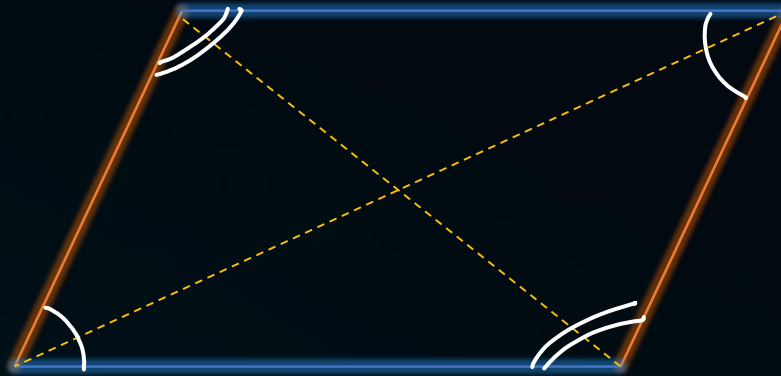
$$2 < c < 12$$

Properties	Parallelogram	Rectangle	Rhombus	Square	
Opp. Sides =	✓	✓	✓	✓	
All Sides =	✗	✗	✓	✓	
Opp. Sides Parallel	✓	✓	✓	✓	
Opp. Angles =	✓	✓	✓	✓	
All Angles = 90	✗	✓	✗	✓	
Diagonals =	✗	✓	✗	✓	
Diagonals bisect each other	✓	✓	✓	✓	
Diagonals bisect each other at 90	✗	✗	✓	✓	
Diagonals bisect Vertex Angles	✗	✗	✓	✓	
Diagonals form four triangles of = area	✓	✓	✓	✓	

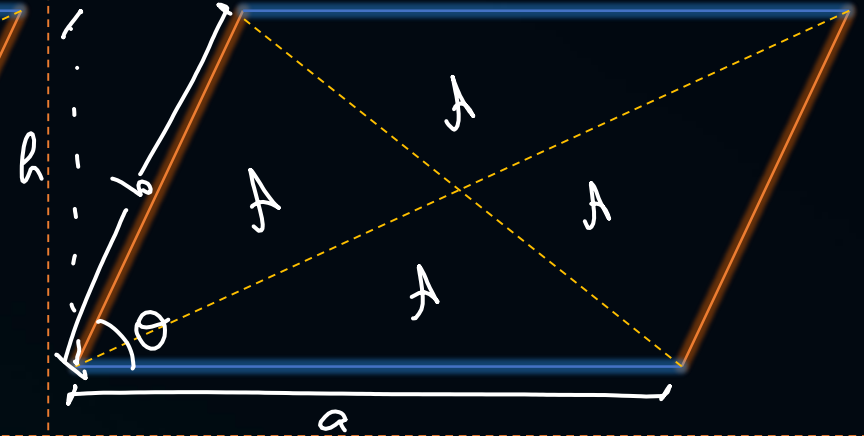
PARALLELOGRAM



opp sides $\parallel$ & equal
diag bisect each
other



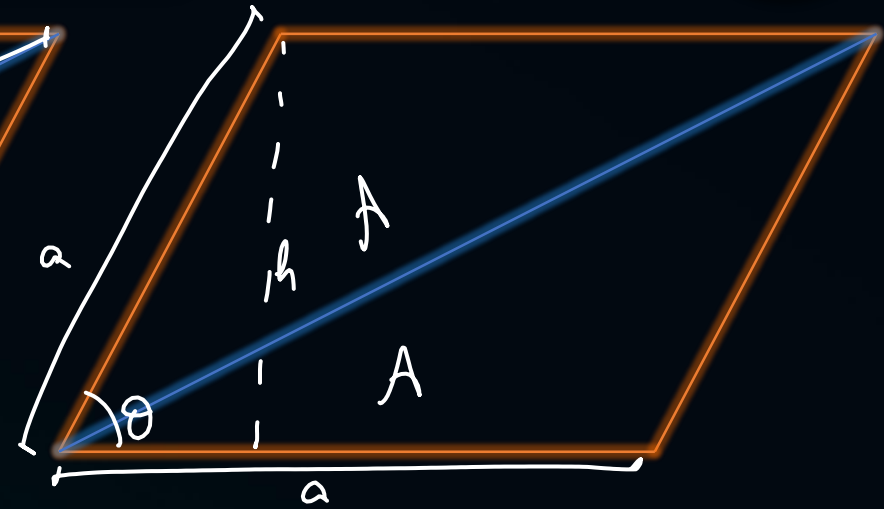
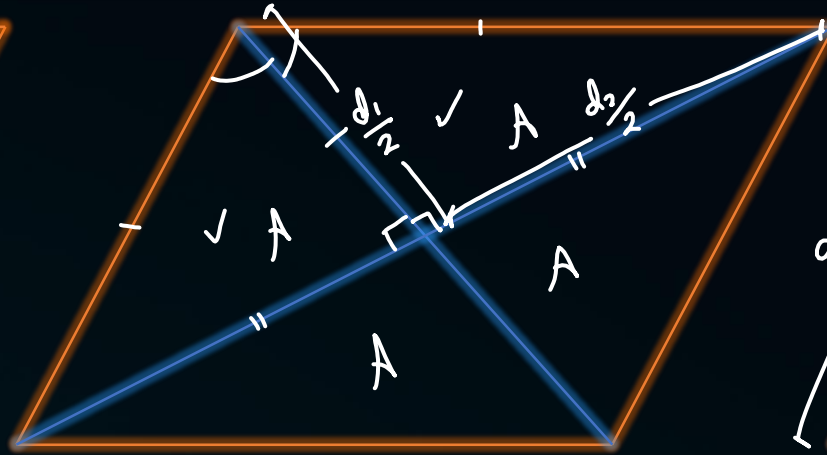
opp $\angle$ s equal
adj. $\angle$ s sum to 180°



$$\text{Area} = \text{base} \times \text{ht} = a(h)$$

$$\text{Area} = ab \sin \theta$$

Rhombus



Sides & Angles

All sides =
Opples =
adjls sum to 180

Diagonals

- * bisect each other
- + Angle bisector
- + bisect at 90

Area

$$\frac{1}{2} d_1 d_2$$

$$a^2 \sin \theta$$

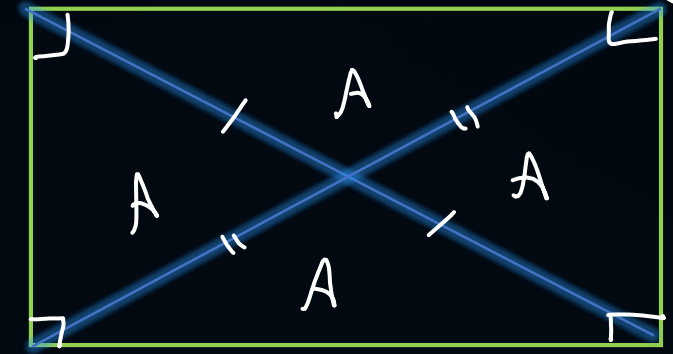
$$ah$$

$$\frac{1}{2} \left(\frac{d_1}{2} \right) \left(\frac{d_2}{2} \right)$$

$$\frac{1}{8} d_1 d_2$$

$$4 \left(\frac{1}{8} d_1 d_2 \right)$$

Rectangle



Sides & Angles

All $\angle$ s

opp sides = $\parallel$

Area

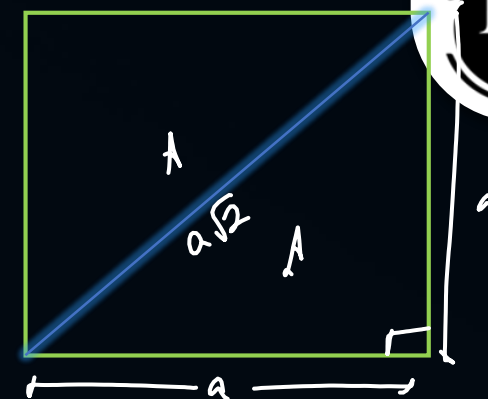
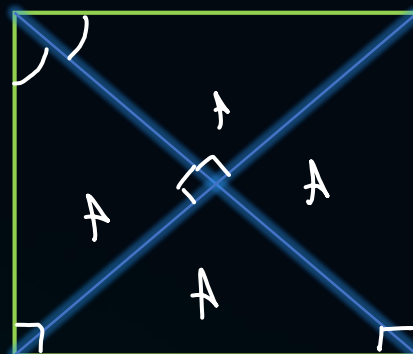
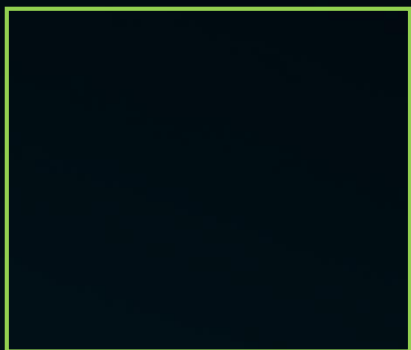
$$A = lb$$

$$2(l+b) = P$$

$$l^2 + b^2 = \text{diag}^2$$

Diagonals

$$\text{Diag} =$$



Sides & Angles

All $\angle$ s 90°

All Sides =

Diagonals

equal

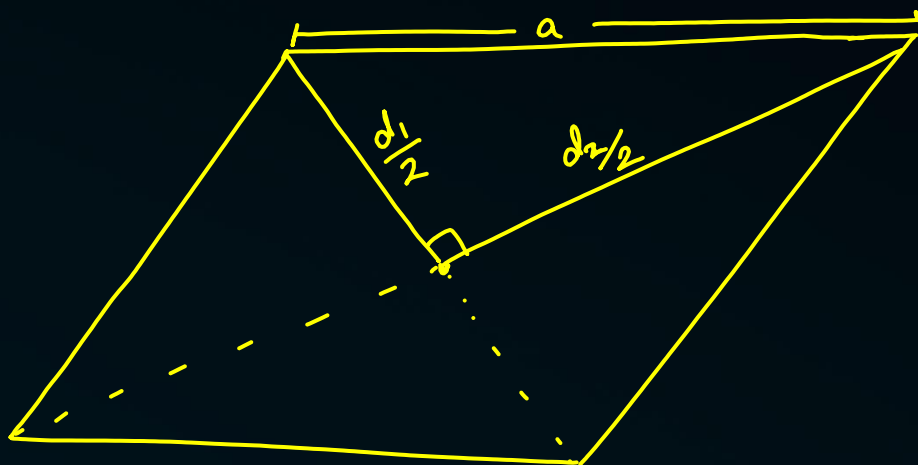
bisect at 90°

$\angle$ bisectors.

Area

a^2

$P = 4a$



$$\text{Perimeter} \checkmark = 4a \checkmark$$

Diag.

$$\text{Area} = \frac{1}{2} d_1 d_2$$

$$\left(\frac{d_1}{2}\right)^2 + \left(\frac{d_2}{2}\right)^2 = a^2$$

$$\frac{d_1^2}{4} + \frac{d_2^2}{4} = a^2$$

$$d_1^2 + d_2^2 = 4a^2$$

$$d_1^2 + d_2^2 = a^2 + a^2 + a^2 + a^2$$

Sum of sq. of diag. = Sum of sq. of sides

QUESTION - 4



#Q. ABCD is a parallelogram. Find the length of Perpendicular DX if Area of Triangle OAB = 10 and AB = 8cm.

A 5 ✓

B 10

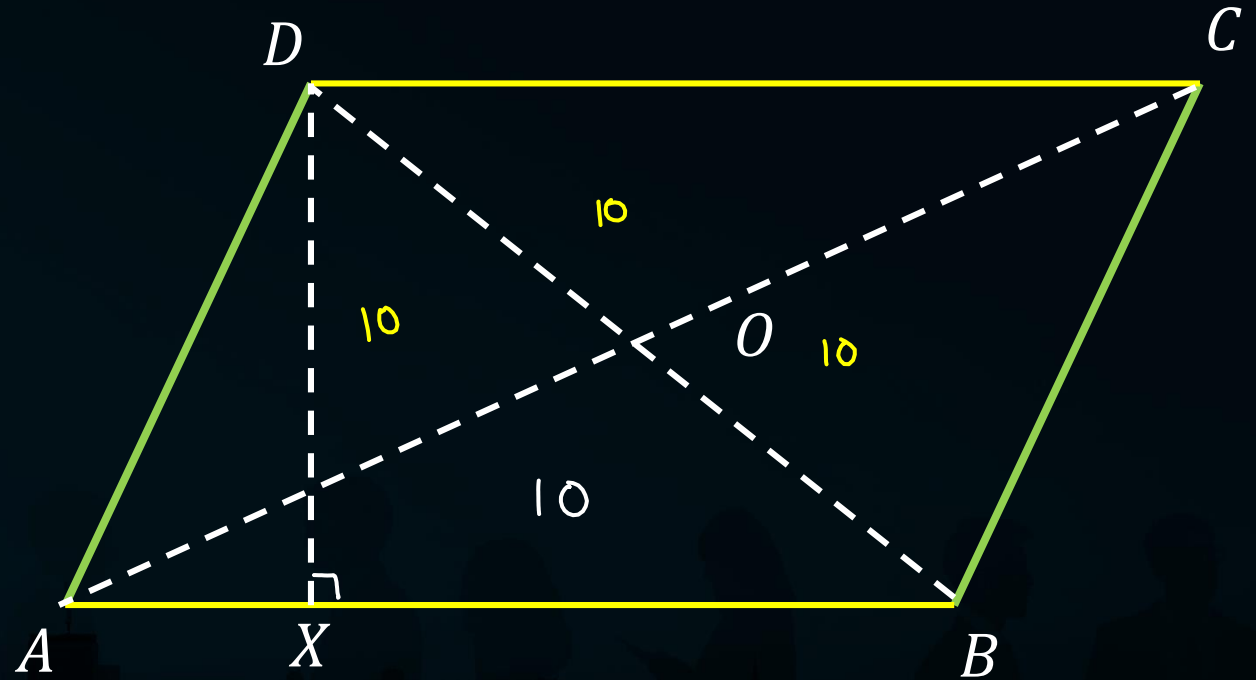
C 15

D 20

$$40 = (AB)(DX)$$

$$40 = 8(DX)$$

$$5 = DX$$



QUESTION - 5



#Q. If the area of a Rhombus is 120 square units and it's one of the diagonal is 10 units, Find the perimeter of the rhombus.

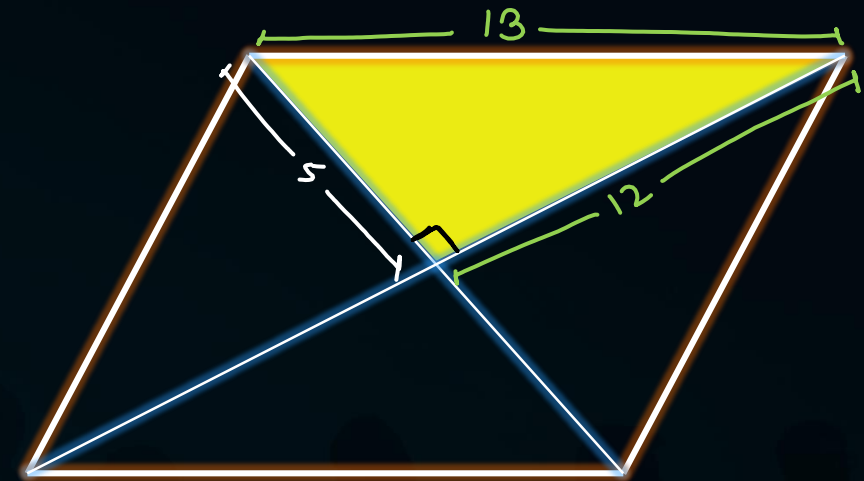
- A** 48
- B** 24
- C** 52 ✓✓
- D** 60

$$A = \frac{1}{2} d_1 d_2$$

$$120 = \frac{1}{2} (10) (d_2)$$

$$d_2 = 24$$

$$P = 13(4) = 52$$



QUESTION – 6 (CAT 2020)



#Q. A circle is inscribed in a rhombus with diagonals 12 cm and 16 cm. The ratio of the area of the circle to the area of the rhombus is

A

$$2\pi/15$$

$$\begin{aligned} \text{Area}_{(\text{RHOMBUS})} &= \frac{1}{2} (12)(16) \\ &= 96 \checkmark \end{aligned}$$

B

$$6\pi/25 \checkmark \checkmark$$

C

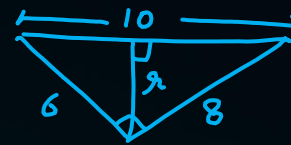
$$3\pi/25$$

$$\frac{\text{Area}_{(\text{CIRCLE})}}{\text{Area}_{(\text{RHOM})}} = \frac{\left[\frac{24}{5} \times \frac{6}{5} \right]}{96}$$

D

$$5\pi/18$$

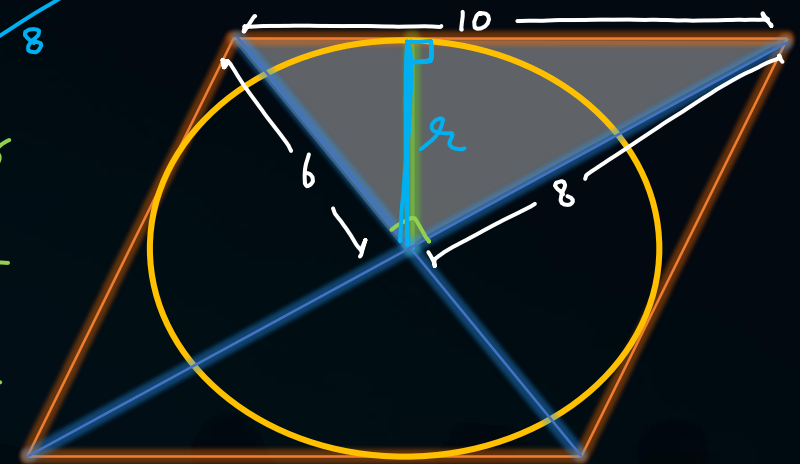
$$\begin{aligned} &= \frac{6\pi}{25} \end{aligned}$$



$$48 = 10 \times$$

$$r = \frac{48}{10}$$

$$r = \frac{24}{5}$$



QUESTION – 7



#Q.

A single quadrilateral, having all four sides equal, is cut out from a rectangular sheet of 36 cm by 24 cm and the left over pieces are scrapped. Find the maximum possible perimeter of all the cutout part that is scrapped.

with (Max POSSIBLE AREA)

A

96

B

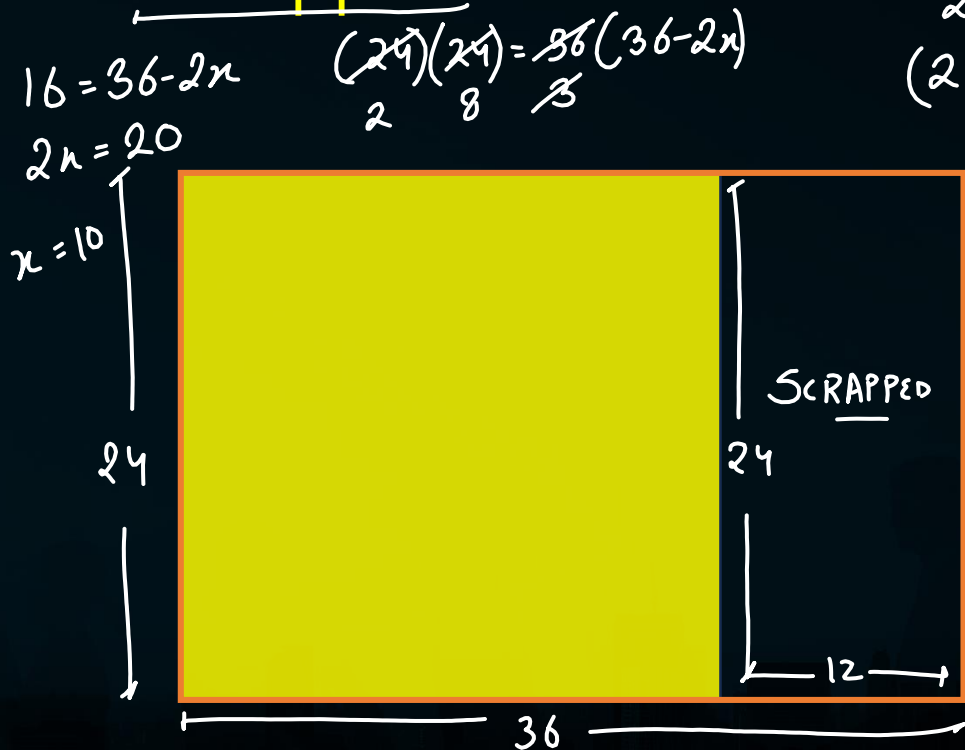
104

C

120 ✓

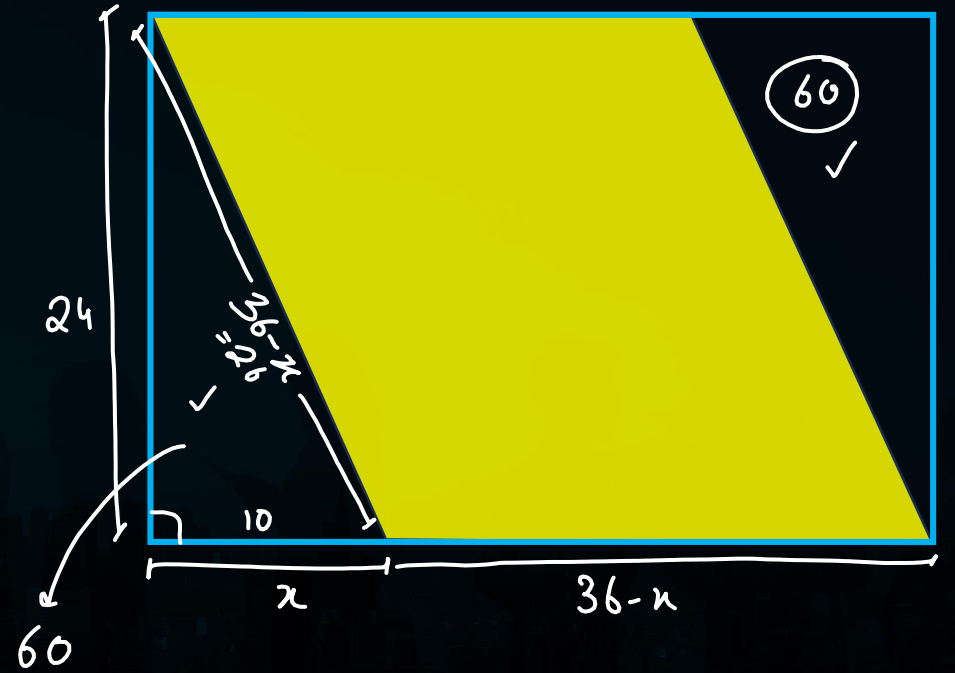
D

72



$$24^2 + x^2 = (36-x)^2$$

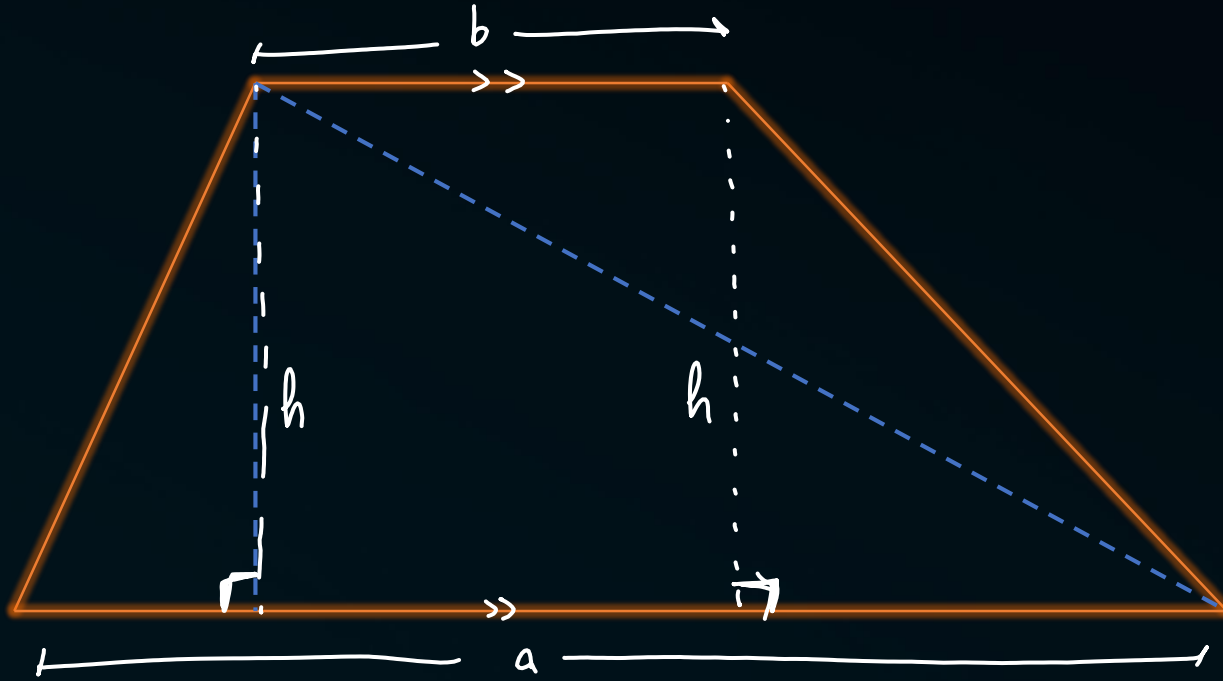
$$(24)(24) = (36-x)^2 - x^2$$





Trapezium

Trapezium



AREA OF TRAPEZIUM =

$$\frac{1}{2} (a+b) h$$

(Avg Base)(ht)

QUESTION – 8

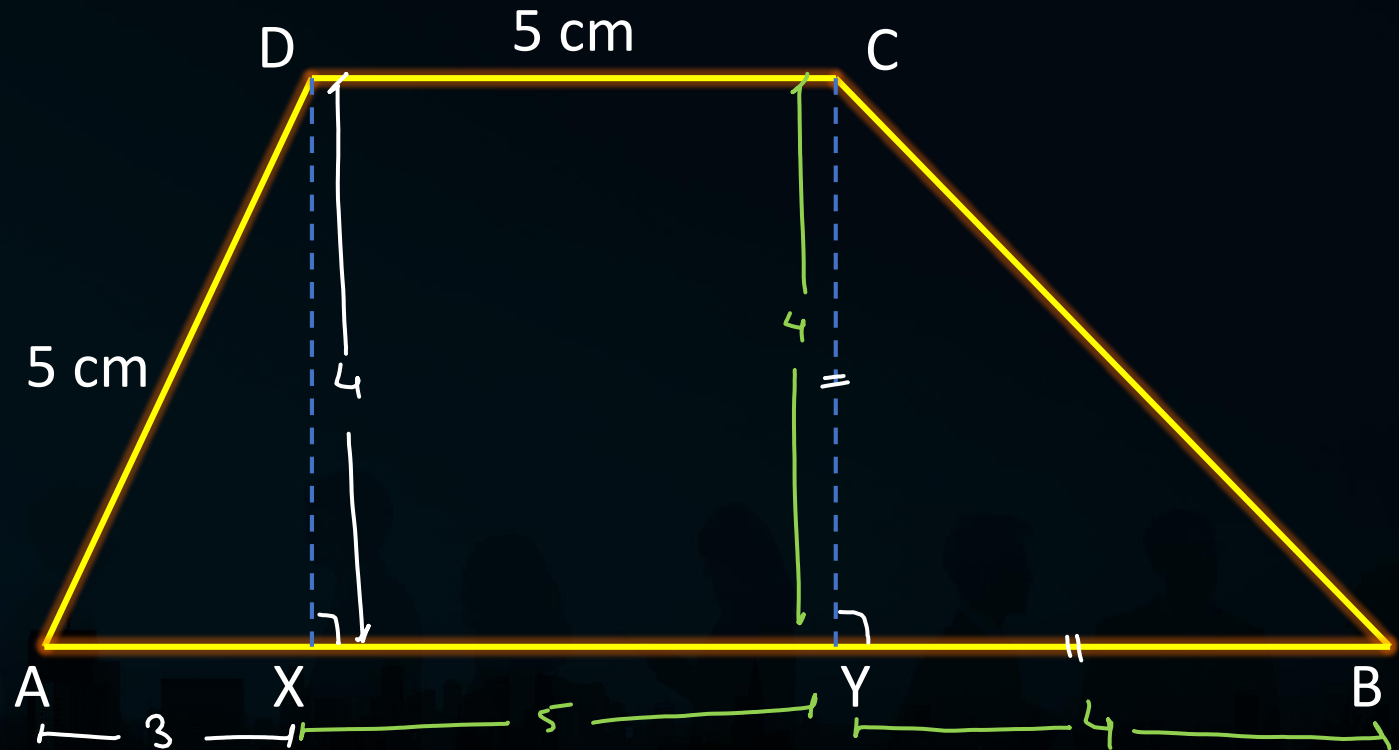


#Q. Find the area of trapezium ABCD if $AD = CD = 5\text{cm}$, $AX = 3\text{cm}$ and $CY = BY$. Given that DX and CY are perpendiculars.

- A** 17
- B** 34 ✓
- C** 51
- D** 68

$$\frac{1}{2}(5+12)(4)^2$$

34

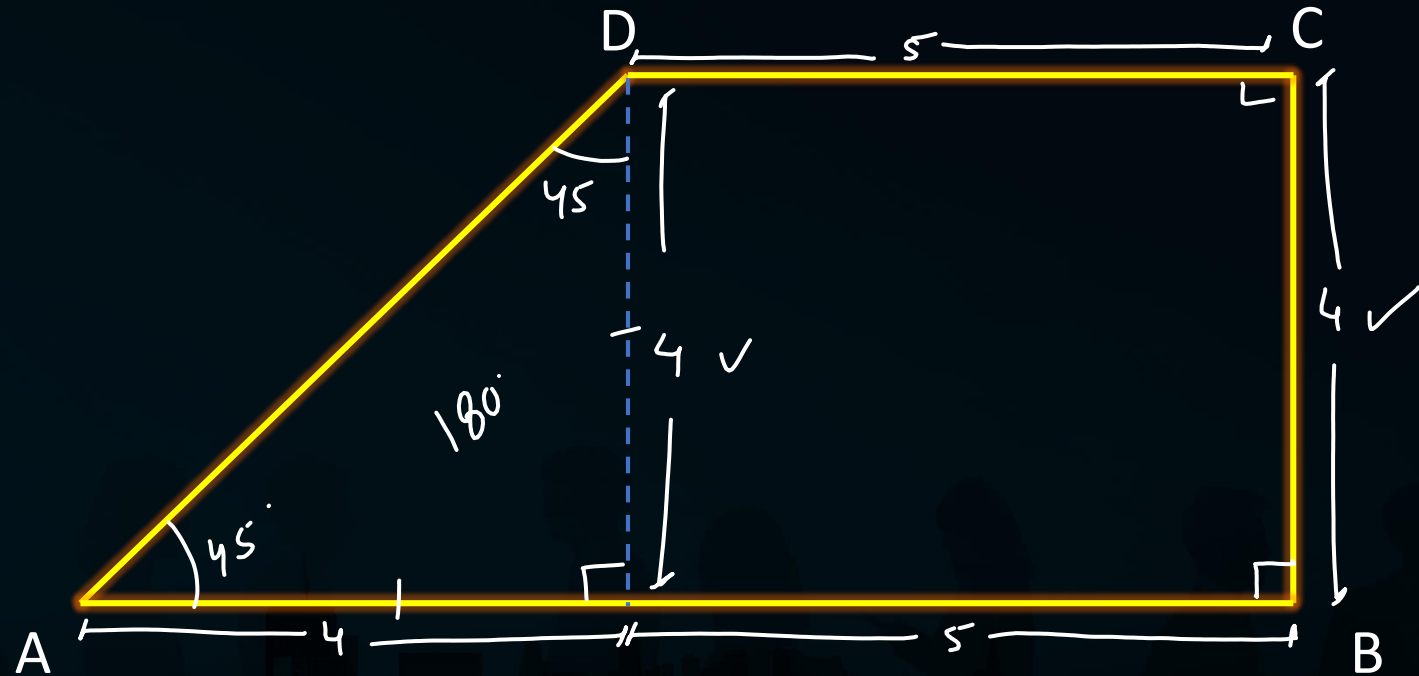


QUESTION – 9 (CAT 2020)

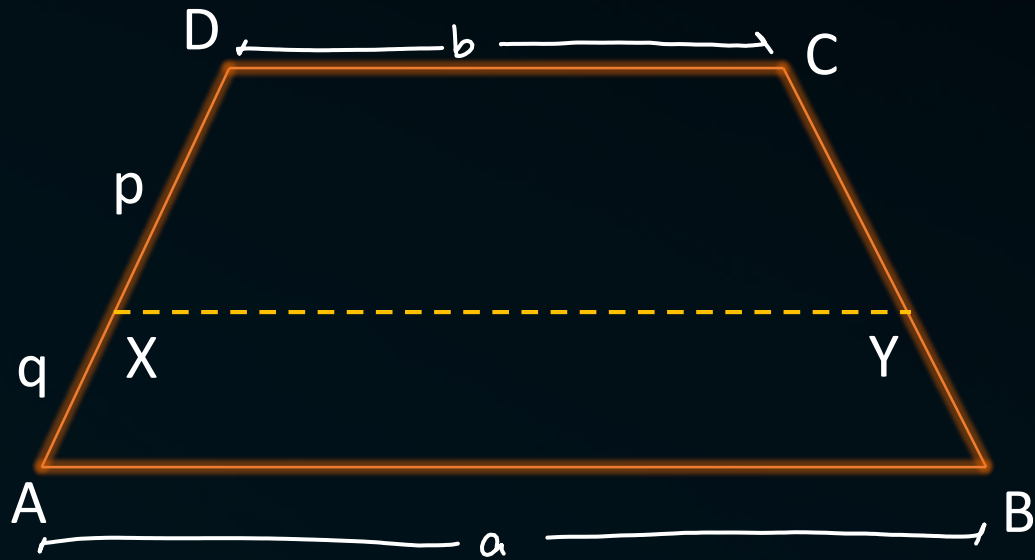


#Q. In a trapezium ABCD, AB is parallel to DC, BC is perpendicular to DC and $\angle BAD = 45^\circ$. If $DC = 5$ cm, $BC = 4$ cm, the area of the trapezium in sq. cm is

$$\frac{1}{2}(5+9) \times 4$$
$$28\checkmark$$



Important Concept



$$XY = \frac{pa + qb}{p + q}$$

QUESTION - 10



#Q. In a trapezium ABCD, EF is Parallel to AB and CD. Find EF if DE: EA = 2:3.

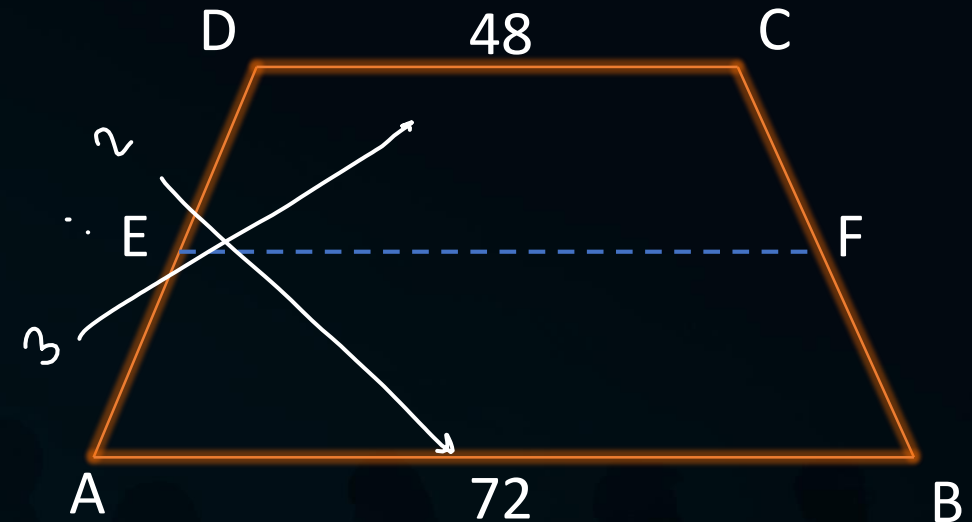
A 56.2

B 50.6

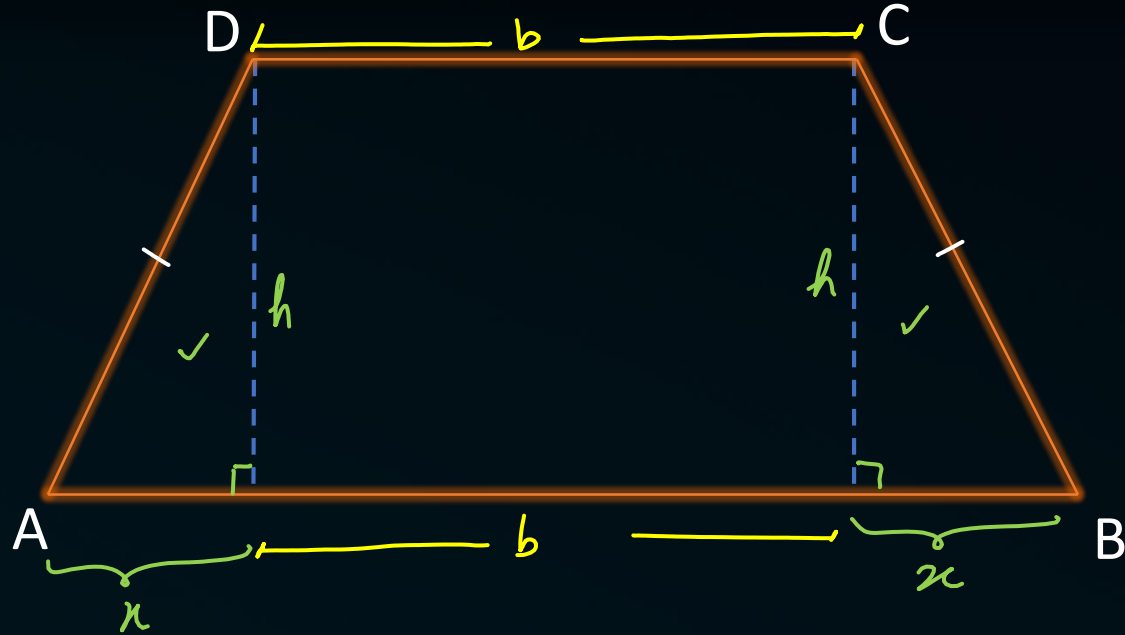
C 57.6 ✓✓

D 63.6

$$\frac{144 + 144}{5} = \frac{288}{5} = 57.6$$



Isosceles Trapezium

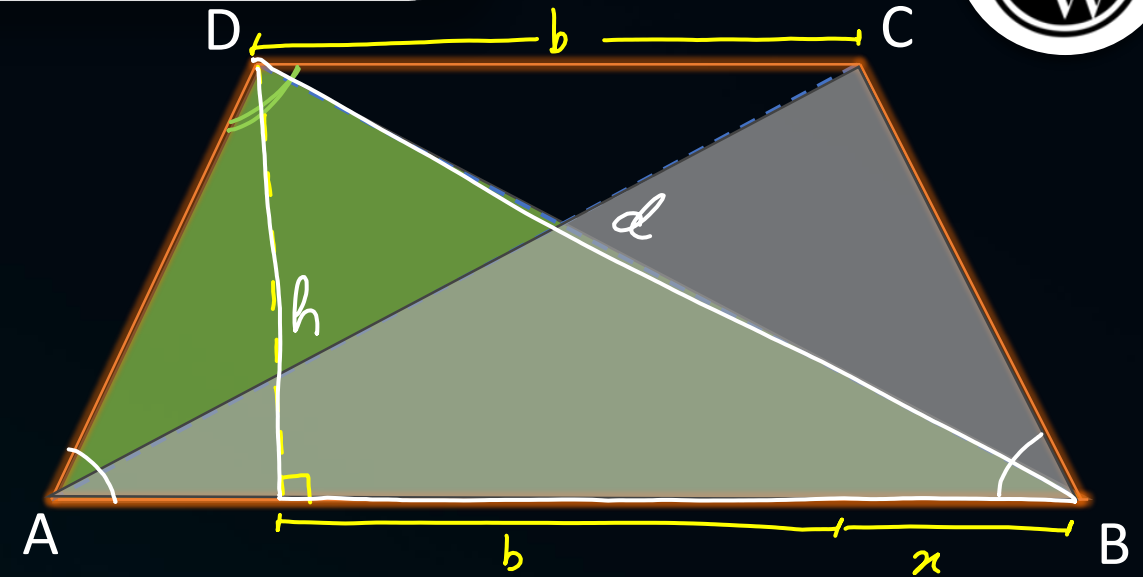


$$\frac{1}{2} (b + b + 2x) h$$

$$\frac{1}{2} (2b + 2x) h$$

$$\frac{1}{2} 2 (b + x) h$$

$$(b + x) h$$



DIAG becomes equal in
isosceles TRAP.

$$h^2 + (b + x)^2 = d^2$$

QUESTION – 11



#Q. The height of isosceles trapezoid ABCD is 24 units. The length of diagonal BD is 25 units. What is the area of trapezoid ABDC?

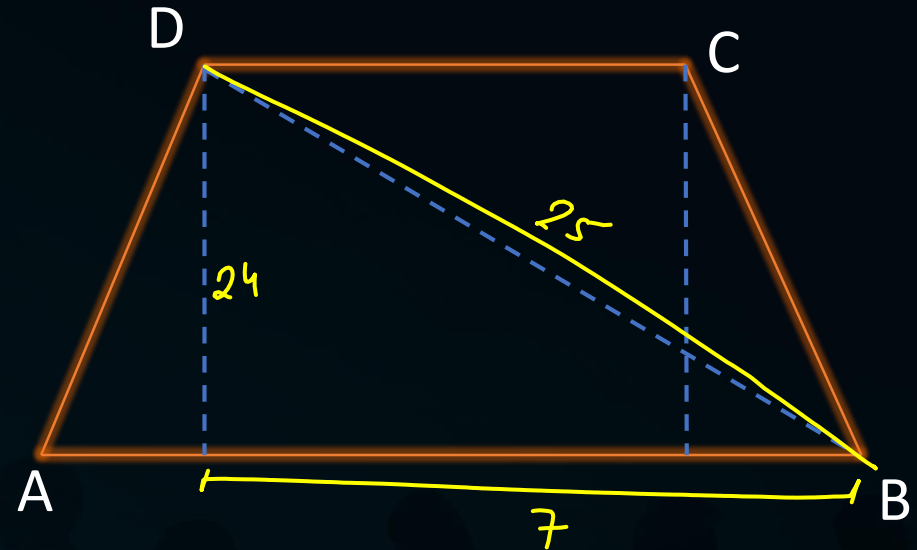
A 84

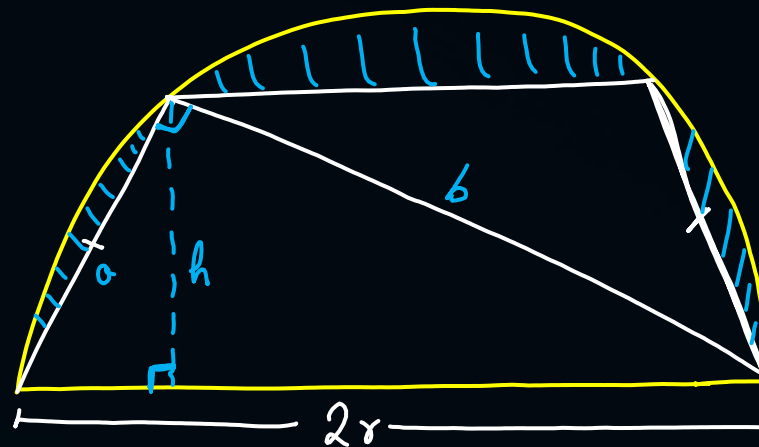
B 96

C 128

D 168 ✓

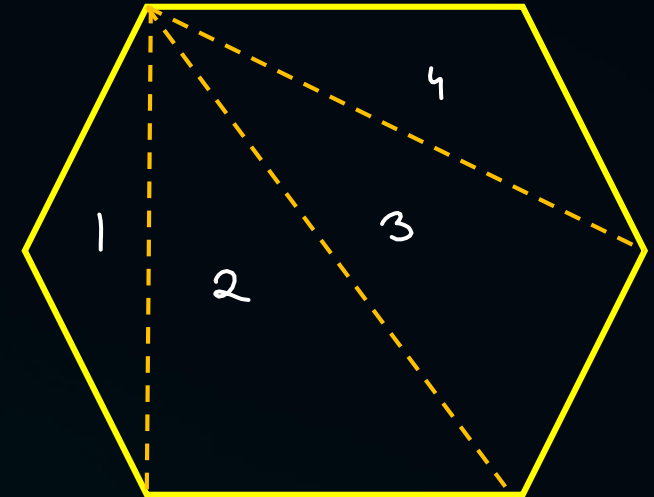
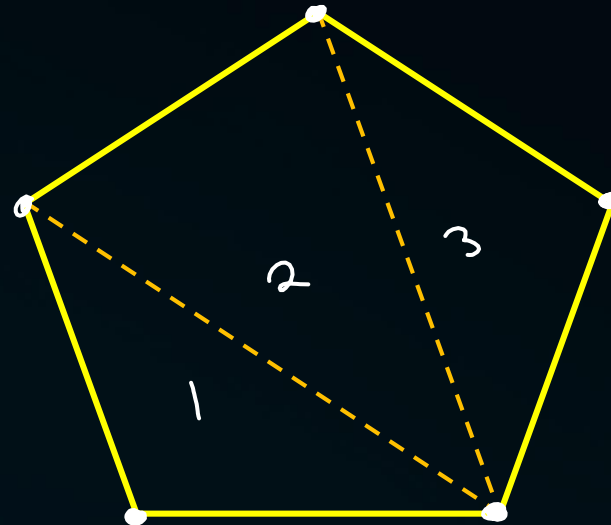
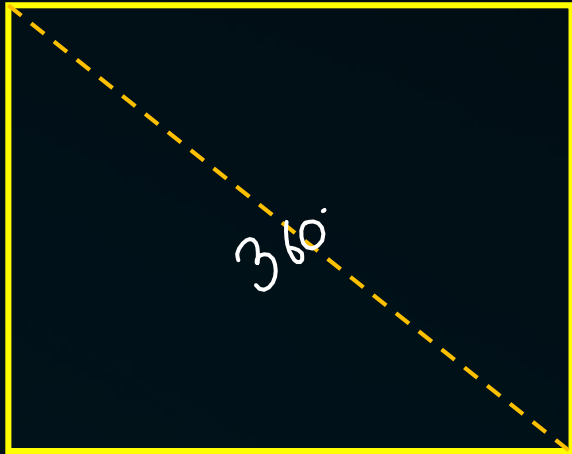
$$24(7) = 168$$





Topic: Polygon

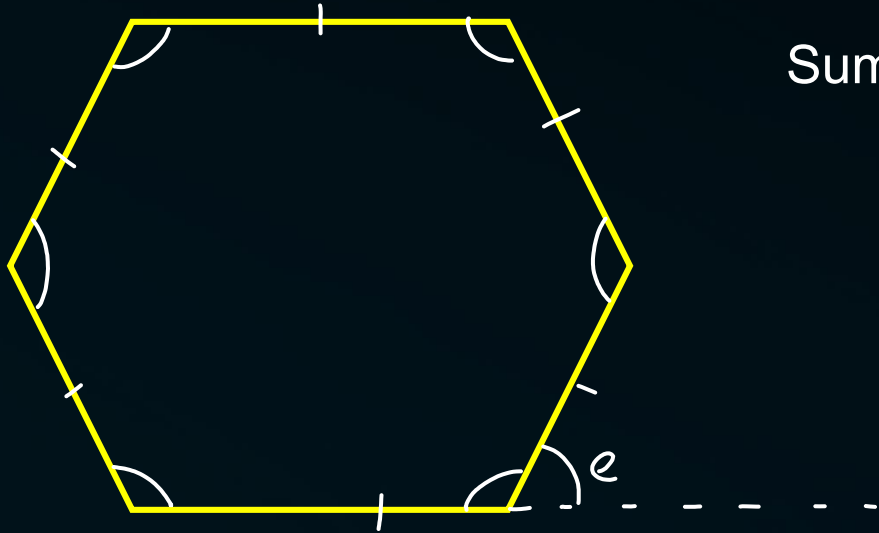
Polygons



$$\text{Sum of Angles} = (n-2) 180^\circ$$

$$\text{Number of Diagonals} = \frac{n(n-3)}{2} \checkmark$$

Regular Polygons



$$\text{Sum of Angles} = (n-2)180^\circ$$

$$\text{Vertex Angle} = \frac{(n-2)180^\circ}{n}$$

$$n = \frac{360^\circ}{e}$$

$$e = \frac{360}{n}$$

CAT 2023
PYQ

if ext angle of
regular polygon = 30° , find it's diag.

$$n = \frac{360}{30} = 12$$

$$d = \frac{6 \times 2(9)}{2} = 54$$

QUESTION – 12



#Q. Find the vertex angle of a regular octagon.

A 120

B 135 ✓

C 160

D 105

$$\begin{array}{r} 3 \quad 45^\circ \\ \underline{(8-2) 180} \\ 8 \end{array}$$

135°

QUESTION – 13 (CAT 2022 Slot 2)



#Q. Regular polygons A and B have number of sides in the ratio 1 : 2 and interior angles in the ratio 3 : 4. Then the number of sides of B equals

A

5

A

n

B

$2n$

B

10 ✓✓

$$\frac{(n-2)180}{n}$$

$$\frac{(2n-2)180}{2n}$$

C

15

D

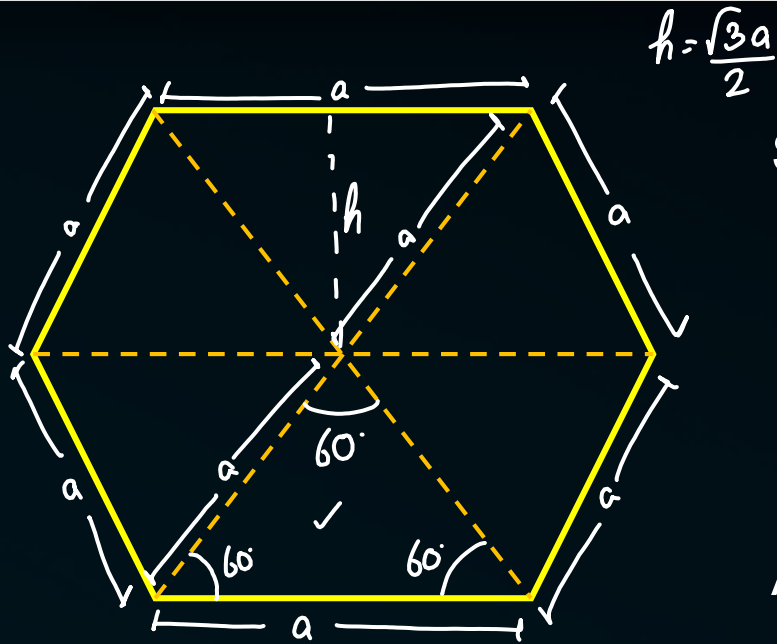
20

$$\frac{n-2}{n-1} = \frac{3}{4}$$

$$4n-8=3n-3$$

$$\boxed{n=5}$$

Regular Hexagon



$$h = \frac{\sqrt{3}a}{2}$$

$$\text{Sum of Angles} = (6-2) \times 180 = 720$$

$$\text{Vertex Angle} = \frac{720}{6} = 120^\circ$$

$$\text{Area} = 6 \left[\frac{\sqrt{3}}{4} a^2 \right]$$

$$\text{Longest Diagonal} = 2a$$



$$\text{Shorter Diagonal} = \sqrt{3}a$$



QUESTION – 14 (CAT 2021 – Slot – 1)



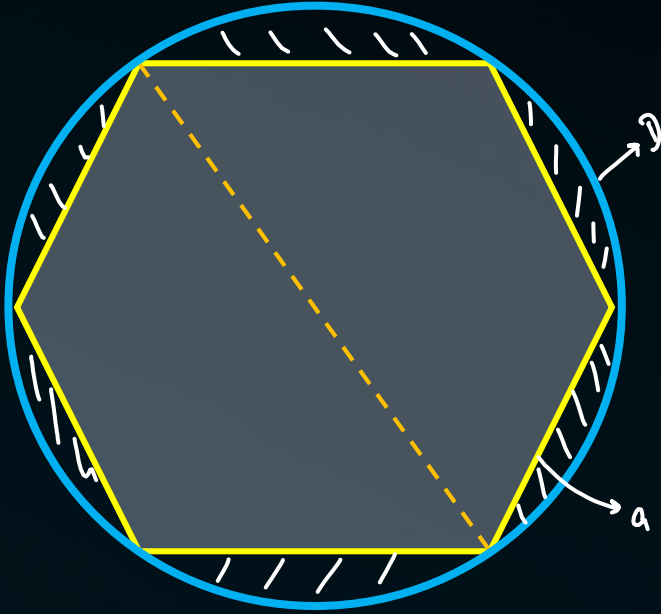
#Q. If the area of a regular hexagon is equal to the area of an equilateral triangle of side 12 cm, then the length, in cm, of each side of the hexagon is

- A $4\sqrt{6}$
- B $2\sqrt{6}$ ✓
- C $\sqrt{6}$
- D $3\sqrt{6}$

$$6 \left[\frac{\sqrt{3}}{4} a^2 \right] = \frac{\sqrt{3}}{4} (12)^2$$

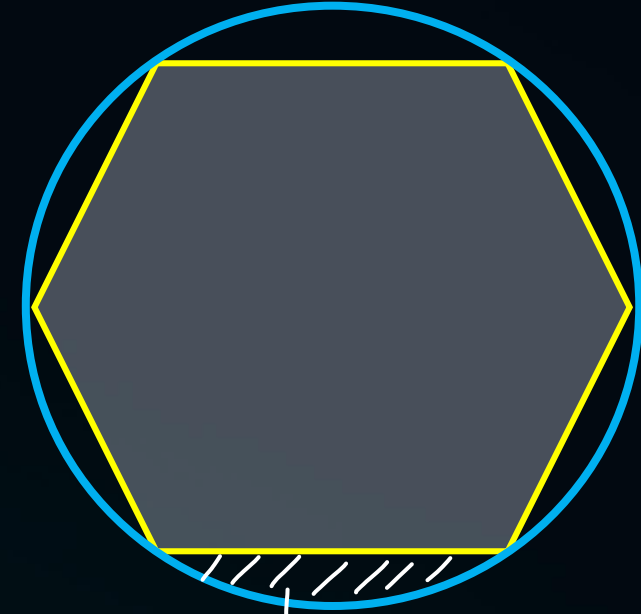
$$a^2 = 24$$

Regular Hexagon and Circles



$$D = 2a$$

$$R = a$$

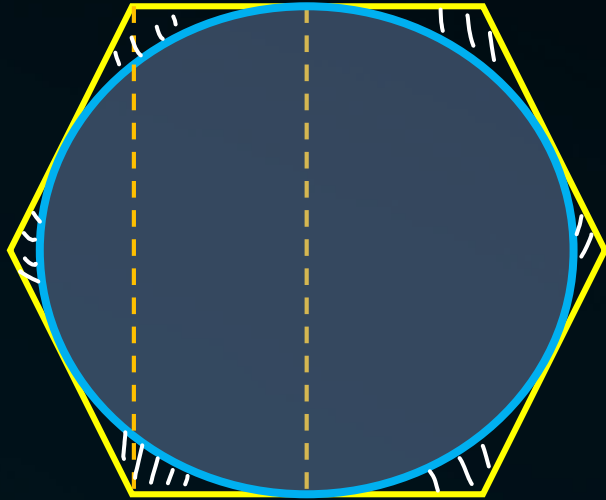


$$\pi R^2 - 6 \left[\frac{\sqrt{3}}{4} a^2 \right]$$

$$\pi a^2 - 6 \left[\frac{\sqrt{3}}{4} a^2 \right]$$

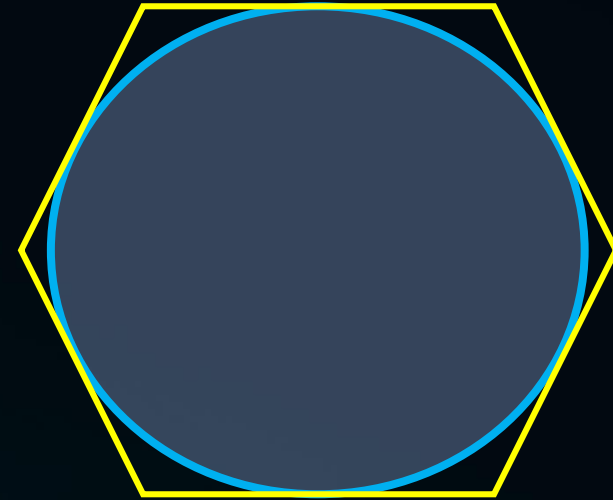
$$\frac{1}{6} [\dots]$$

Regular Hexagon and Incircle



$$D = \sqrt{3} a$$

$$r = \frac{\sqrt{3} a}{2}$$



$$6 \left[\frac{\sqrt{3}}{4} a^2 \right] - \pi r^2$$

$$6 \left[\frac{\sqrt{3}}{4} a^2 \right] - \pi \left[\frac{\sqrt{3} a}{2} \right]^2 = 6 \left[\frac{\sqrt{3}}{4} a^2 \right] - \pi \left(\frac{3 a^2}{4} \right)$$



THANK
You

